Lesson 10: Play It Again

001 Game Creator (Windows only)

VIDEO GAME CREATION

This week: when the round ends, the game actually stops - and pressing R brings it back to a genuinely fresh start.

Open your own [YourName]Game project from Lesson 9 - don't start a new one.

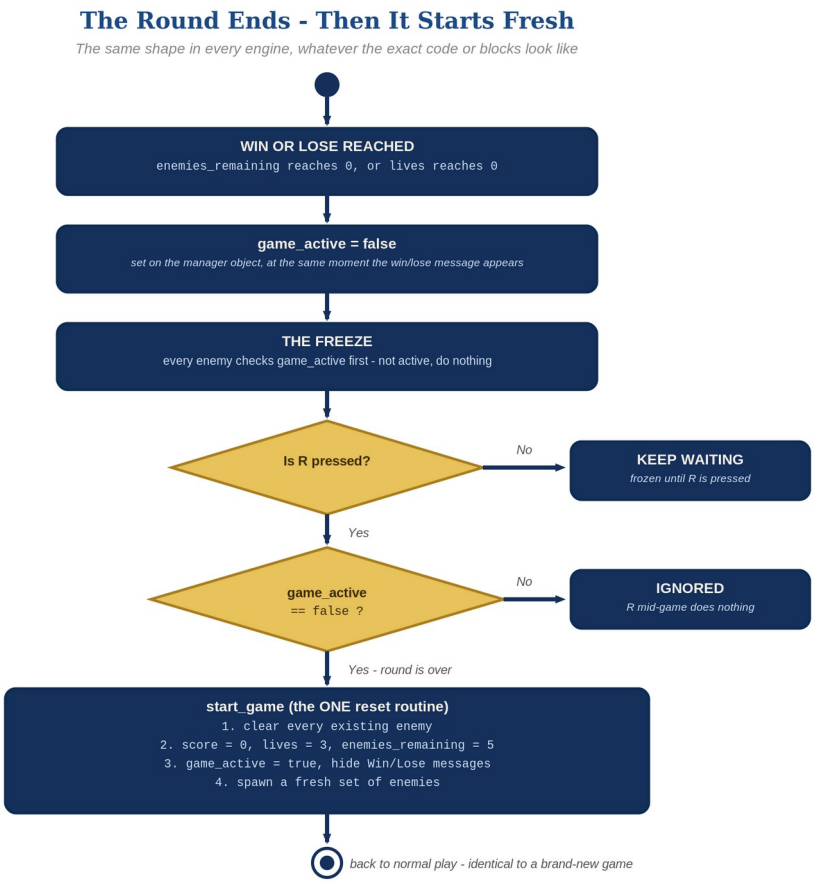
Tip: no new assets this week - this is a logic lesson, and it reuses your very first setup code from Lesson 5.

Setup

1. Open your [YourName]Game project from Lesson 9.
2. Create a new Variable (Resources > Variables/Switches/Collections/Tables) named game_active, default value 1.
3. Create a new Custom Event (Resources > Custom Events) named ResetGame - this is 001 Game Creator's reusable-subroutine feature, callable from any trigger's script.
4. Create a new Input Trigger for the R key (Input Set > Add Control Combination > press R > choose "Pressed" > Edit Script) - the same underlying mechanism your movement and shoot controls have used since Lesson 2.

Level 1 - Minimal (MVP)

When the round ends (win or lose), the game freezes - score and lives stop changing, and further enemy hits/escapes do nothing. Pressing R resets everything and starts a new round.



When the round ends, everything freezes - and R, pressed only while the round is over, runs the one reset routine. The same shape in every engine.

5. On the enemy Actor Template's "Collided with actor" trigger, wrap the score/enemies_remaining/win-check actions in an If(game_active = 1) check, and when enemies_remaining reaches 0, also set game_active to 0 (code below).
6. On the enemy Actor Template's "Left/Entered Camera" trigger, wrap the lives-decrement action in the same If(game_active = 1) check, and when lives reaches 0, also set game_active to 0.
7. Update both Message Box texts to include the restart hint: "You Win! - Press R to Restart" and "Game Over - Press R to Restart".
8. On your new Input Trigger (R, Pressed), add: If(game_active = 0) -> Call Custom Event(ResetGame).
9. Build the ResetGame Custom Event: reset score/lives/enemies_remaining/game_active to their starting values, then revive and reposition each of your enemy copies back to their original starting spots (code below).
10. Save, then Play/Test: play to a win or loss, then press R and confirm everything resets and play resumes.

```
enemy Actor Trigger: Collided with actor (revised) ->
  If(game_active = 1)
    Variable – Modify Value(score, Add, 10)
    Variable – Modify Value(enemies_remaining, Subtract, 1)
    If(enemies_remaining <= 0)
      Variable – Modify Value(game_active, Set, 0)
      Message Box("You Win! - Press R to Restart")
    Actor – Destroy/Revive Actor(relate, Destroy)
    Actor – Destroy/Revive Actor(this, Destroy)

enemy Actor Trigger: Left/Entered Camera (revised) ->
  If(game_active = 1)
    Variable – Modify Value(lives, Subtract, 1)
    If(lives <= 0)
      Variable – Modify Value(game_active, Set, 0)
      Message Box("Game Over - Press R to Restart")
    Actor – Destroy/Revive Actor(this, Destroy)

Input Trigger: R (Pressed) ->
  If(game_active = 0)
    Call Custom Event(ResetGame)

Custom Event: ResetGame ->
  Variable – Modify Value(score, Set, 0)
  Variable – Modify Value(lives, Set, 3)
  Variable – Modify Value(enemies_remaining, Set, 5)
  Variable – Modify Value(game_active, Set, 1)
  Actor – Destroy/Revive Actor(enemy_1, Revive) -> Actor –
Teleport(enemy_1, [starting position])
  (repeat for enemy_2, enemy_3, enemy_4, enemy_5)
```

Level 2 - Complete (Alpha/Beta)

A restart is a genuinely clean slate - no leftover enemies, no stale score/lives, and it works correctly no matter how many times in a row you play.

- Play to a win, then press R and confirm score/lives/enemies_remaining are exactly back to 0/3/5, with all 5 enemy copies revived at their starting positions.
- Repeat for a loss - same clean reset.
- Press R while the game is still active and confirm NOTHING happens - the If(game_active = 0) check on your Input Trigger is what prevents an accidental reset.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Visual: fade the screen briefly (a simple Camera or Interface fade effect, if you're comfortable exploring the wiki further) the moment game_active flips to 0.
- Structure: notice ResetGame is now the single place that sets up a fresh game, callable from your Input Trigger any time it's needed - the same setup values (score, lives, enemies_remaining) only ever need to be written once.
- Design note: because your enemies use Automatic Movement rather than a spawn loop, this lesson intentionally doesn't force-stop enemies already in motion at the moment of game-over - they'll simply drift off naturally within a second or two, harmlessly, since the If(game_active = 1) check means they no longer affect score or lives once the round has ended. This is a deliberate, reasonable simplification for this engine, not a cut corner.

If Something's Wrong

- Restart does nothing when you press R: confirm the Input Trigger's Control Combination is actually set to the R key and "Pressed" (not "Released"), and that game_active is actually 0 at that point.
- Score/lives keep changing after "Game Over": confirm the If(game_active = 1) check was actually added to BOTH triggers (Collided with actor AND Left/Entered Camera), not just one.
- Enemies don't reappear after restart: confirm each enemy copy's exact name (enemy_1, enemy_2, etc.) matches what you use in the ResetGame Custom Event - a typo'd name silently does nothing.

Want More?

001gamecreator.com/wiki/customevents/, 001gamecreator.com/wiki/inputtriggers/, Student Resource List > 001 Game Creator